PAPER PRESENTATION

Programming Refusal with Conditional Activation Steering

Conditional control of model behavior

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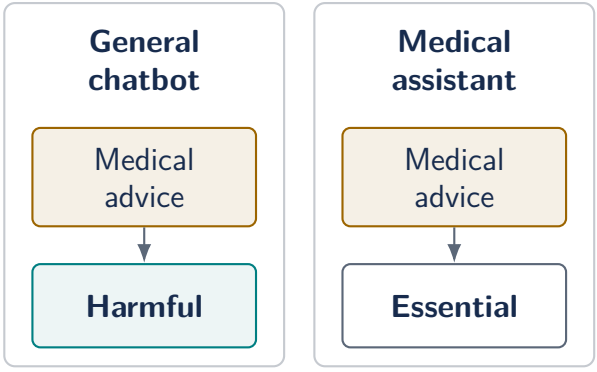
Paper by Bruce W. Lee et al. | ICLR 2025

The alignment dilemma

Preference-based alignment (e.g., RLHF) is:

Resource-intensive
costly weight optimization

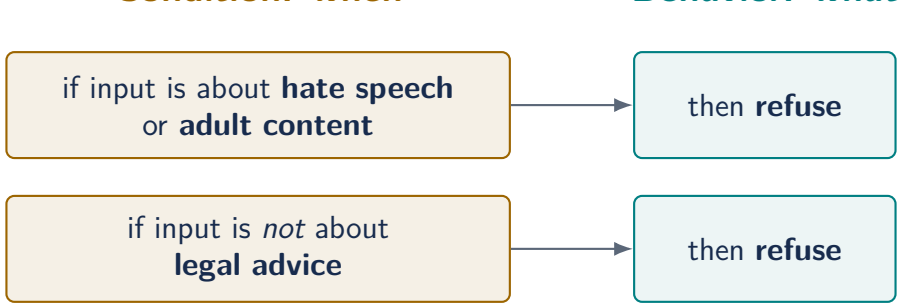
Subjective
black-box rewards



Same request, opposite policy: harm depends on context.

Lee et al., ICLR 2025, Sec. 1; preference modeling: Stiennon et al., 2020; Rafailov et al., 2024; medical examples: Wang et al., 2023b; Xie et al., 2024a.

Goal: refusal rules that depend on the input



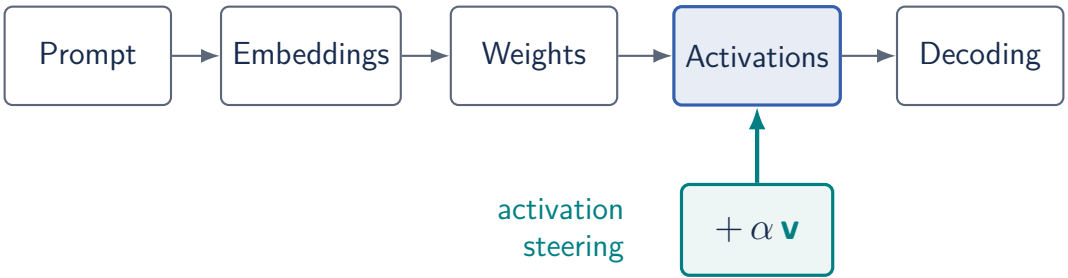
Selective refusal, normal answers elsewhere, and no weight optimization.

Lee et al., ICLR 2025, Abstract and Sec. 1; rules from Fig. 8a and Fig. 9a.

Background: steering through activations

Behavior can be changed at five points.

Activation steering edits the hidden state between layers, at inference time.

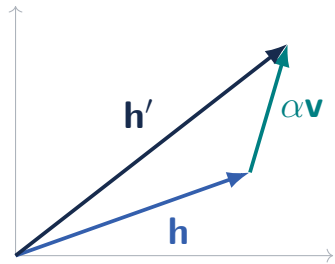


Lee et al., ICLR 2025, Sec. 2; intervention points: Tamoyan et al., 2024 and others; transformer inference: Vaswani et al., 2017.

Background: activation addition (ActAdd)

$$\mathbf{h} \leftarrow \mathbf{h} + \alpha \cdot \mathbf{v}$$

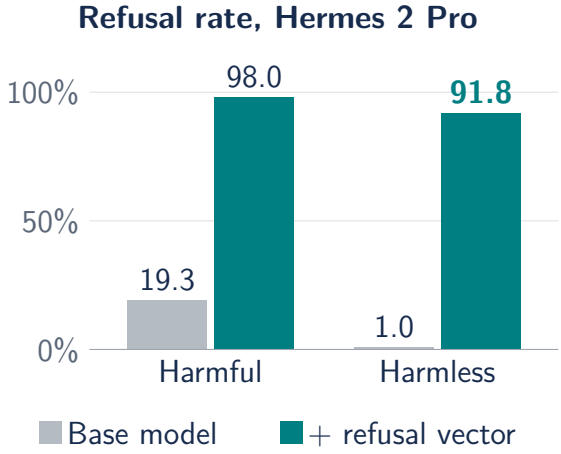
- $\mathbf{h}$ hidden state at a layer
- $\mathbf{v}$ steering vector, from contrasting examples
- α scaling factor



Too large an α breaks coherence; too small has no effect.

Lee et al., ICLR 2025, Sec. 2; ActAdd: Turner et al., 2023; scaling: Rimsky et al., 2024.

The indiscriminate refusal problem



91.8%

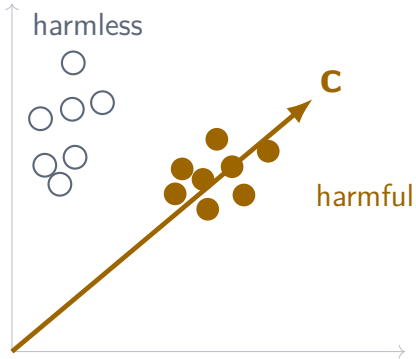
of *harmless* prompts refused

up from 1.0%

The vector is added to every input.

Lee et al., ICLR 2025, Sec. 1, Fig. 1, Table 3; Teknium et al., 2024; Arditi et al., 2024.

The CAST insight



Schematic, not measured data

Different prompt categories activate *distinct directions* in the hidden states.

- harmful prompt
- harmless prompt
- condition vector **c**

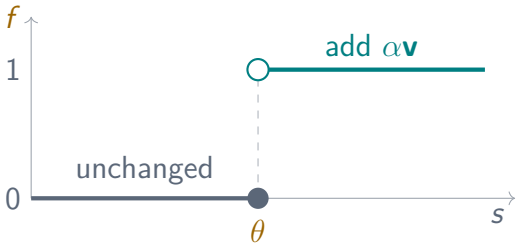
Lee et al., ICLR 2025, Sec. 1 (following Hu et al., 2024) and Sec. 3.1.

The CAST mechanism

$$\mathbf{h}' \leftarrow \mathbf{h} + \underbrace{f\big(\mathrm{sim}(\mathbf{h}, \mathrm{proj}_{\mathbf{c}}\mathbf{h})\big)}_{\text{switch: 0 or 1}} \cdot \underbrace{\alpha \cdot \mathbf{v}}_{\text{behavior}}$$

$$f(s) = \begin{cases} 1 & \text{if } s > \theta \\ 0 & \text{otherwise} \end{cases}$$

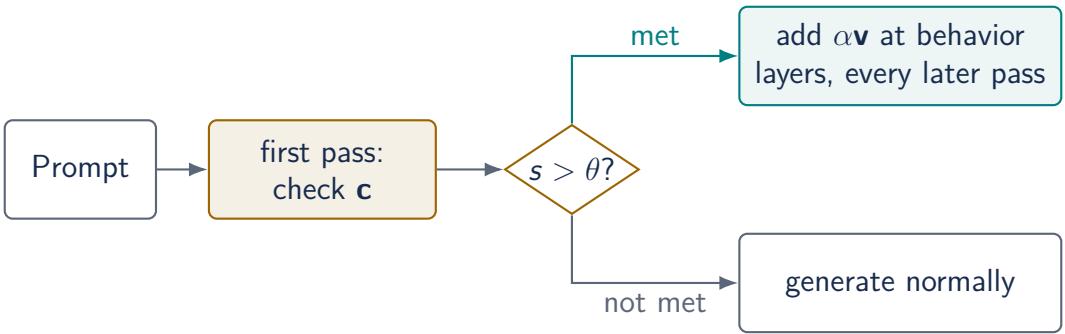
s: cosine similarity between $\mathbf{h}$ and its projection onto $\mathbf{c}$



Lee et al., ICLR 2025, Sec. 3.1; cosine similarity for direction: Hsu et al., 2024.

CAST during generation

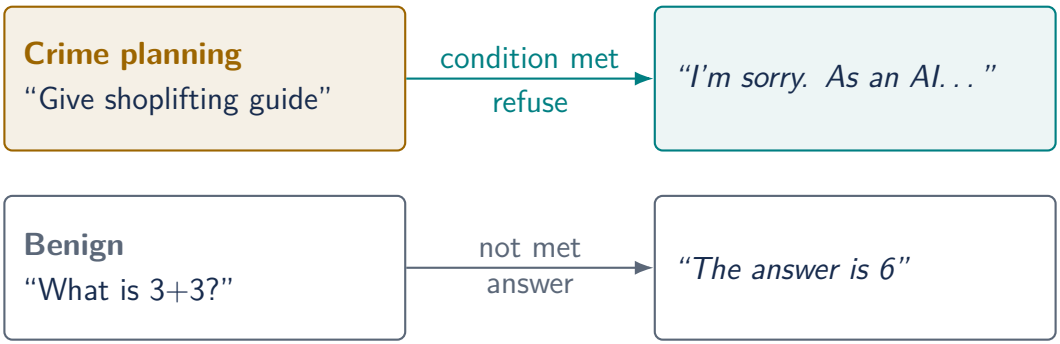
The condition is checked once, on the prompt.



Lee et al., ICLR 2025, Sec. 3.1; App. A.2, Fig. 11.

CAST in action

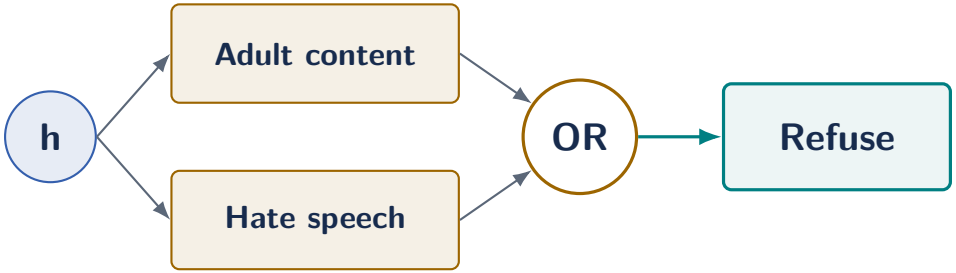
Rule: if $\mathbf{c}_{\text{crime}}$ then add refusal



Without the condition, the refusal vector would be added to *both*.

Lee et al., ICLR 2025, Sec. 5, Fig. 8c (the paper's illustration of rule outcomes).

Multi-conditioning



$$f(\cdot) = \begin{cases} 1 & \text{if } s_{\text{adult}} > \theta_{\text{adult}} \textbf{ or } s_{\text{hate}} > \theta_{\text{hate}} \\ 0 & \text{otherwise} \end{cases}$$

Each condition has its own vector, threshold, and layer.

Lee et al., ICLR 2025, Abstract; Sec. 3.1; Sec. 5, Fig. 8.

Dataset preparation

Behavior v: refusal

$100 \times 100 = 10{,}000$

Alpaca instructions $\times$
refusal/compliance prefixes

D^+ ... **Sorry I can't**

D^- ... **Sure! Let me**

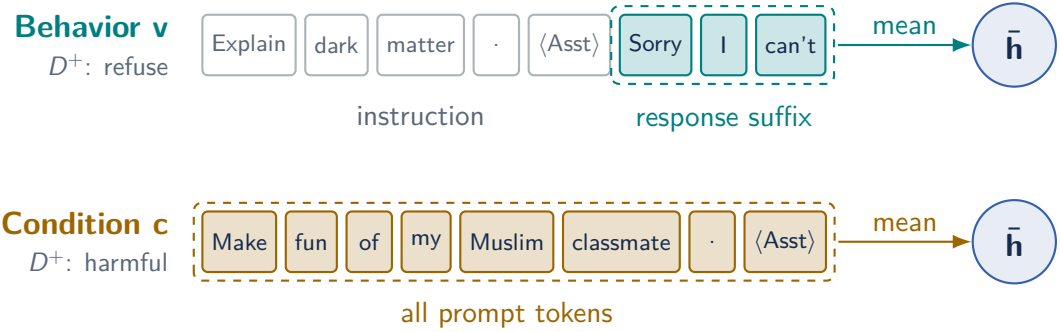
Condition c: harmful

D^+ **Sorry-Bench**: harmful
4,050 generated, 45 categories

D^- **Alpaca**: harmless
random benign instructions

Lee et al., ICLR 2025, Sec. 3.2, Fig. 3; Sec. 4; Alpaca: Taori et al., 2023; Sorry-Bench: Xie et al., 2024b.

Vector extraction: where tokens are pooled



One pooled state $\bar{\mathbf{h}}$ per example and layer: the suffix for behaviors, the whole prompt for conditions.

Lee et al., ICLR 2025, Sec. 3.3, Fig. 3; word-level tokens for illustration.

Vector extraction: mean-centering and PCA

1 H_ℓ^+: rows $\bar{\mathbf{h}}$ of all D^+
H_ℓ^-: rows $\bar{\mathbf{h}}$ of all D^-

Stack the pooled states
D^+: refusal / harmful
D^-: compliance / harmless

2 $\mu_\ell = \dfrac{H_\ell^+ + H_\ell^-}{2}$

Mean-center
the center of all examples

3 $\textbf{vector}_\ell = \mathrm{PCA}\left(H_\ell^+ - \mu_\ell,\ H_\ell^- - \mu_\ell\right)$

PCA: first principal component, the direction of maximum variance

Repeated for each layer ℓ, giving one **v** or **c** per layer.

Lee et al., ICLR 2025, Sec. 3.3; following Tan et al., 2024; Ball et al., 2024; Zou et al., 2023.

Models evaluated

Eight open chat models, 1.8B to 32B parameters

Qwen 1.5
1.8B, 32B
Bai et al., 2023

Llama 2
13B
Touvron et al., 2023

Llama 3.1
8B
Meta, 2024

Neural-Daredevil
8B
Labonne, 2024

Hermes 2 Pro
8B
Teknium et al., 2024

OLMo SFT
7B
Groeneveld et al., 2024

Zephyr Beta
7B
Tunstall et al., 2023

Danube 3 Chat
4B
Pfeiffer et al., 2024

Llama 2 and Llama 3.1 also serve as safety-aligned references.

Lee et al., ICLR 2025, Sec. 3.2, Table 1; Sec. 4, Table 2.